

SECTION A [25 marks]

This section consists of **3** questions. Answer **all** questions.

1. (a) Evaluate the following limits.

(i) $\lim_{x \rightarrow +\infty} \frac{\sqrt{x} + 1}{x - 1}.$ [3 marks]

(ii) $\lim_{x \rightarrow 3^-} \frac{|6 - 2x|}{x^2 - 2x - 3}.$ [3 marks]

(b) Given that $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = 9$, evaluate $\lim_{x \rightarrow c} 2f(x) \bullet \sqrt{g(x)}.$ [3 marks]

2. Given that $y = \sin(\ln x)^2$, show that $x^2(\ln x) \frac{d^2 y}{dx^2} + x(\ln x - 1) \frac{dy}{dx} + 4y(\ln x)^3 = 0.$

[9 marks]

3. The surface area of a sphere is increasing at a rate of $4 \text{ cm}^2\text{s}^{-1}$. Find the rate of increase of the radius when the surface area is $64\pi \text{ cm}^2$.

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. Given that $u = 3i + \sqrt{5}$. Solve the equation below to find the complex number w , where $i = \sqrt{-1}$.

$$10 - u\bar{u} + \frac{3}{2}i = w$$

Hence, find $\arg(\bar{w} - i^5)$. [6 marks]

2. (a) Solve the simultaneous equations:

$$x - 3y = 0$$

$$\log_x 3(1 - \log_3 y) = 1$$

[5 marks]

- (b) Find the set of values of x which satisfies the following inequalities.

(i) $\frac{3 - 2x^2 - x}{x + 3} \geq 0$ [4 marks]

(ii) $|x - 2| < \frac{1}{2}(4 - x)$ [3 marks]

3. The functions g and h are given by

$$g(x) = x^2 + 2kx + 4, x \in \mathbb{R}$$

$$h(x) = 3 - kx, x \in \mathbb{R}$$

where k is a non-zero constant.

- (a) Find, in terms of k , the range of g . [2 marks]

- (b) Given further that $gh(2) = 4$, determine the value of k . [4 marks]

4. The functions f and g are defined by

$$f : x \mapsto \ln(3x - 2), x > \frac{2}{3}$$

$$g : x \mapsto \frac{3}{x - 2}, x \neq 2$$

(a) Find the exact value of $fg(3)$. [2 marks]

(b) Find an expression for $f^{-1}(x)$ and state its domain. [4 marks]

(c) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same diagram. [3 marks]

(d) Find the exact values of $|g(x)| = 4$. [3 marks]

5. (a) Given $g(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$

Determine whether $g(x)$ is continuous at $x = 0$. State your reason. [4 marks]

(b) Function f is defined as follows:

$$f(x) = \frac{x^2}{x^2 - 4}$$

(i) Find all asymptotes of f and the coordinate of point at which the curve intersects the axes. [7 marks]

(ii) Sketch the graph of $f(x)$. [3 marks]

6. (a) A curve is given by the parametric equations $x = \frac{2t}{1 - 2t}$, $y = \frac{t^2}{1 - 2t}$.

Find an expression for $\frac{dy}{dx}$ in terms of t . Hence, find the value of t for which $\frac{dy}{dx} = -2$.

[7 marks]

(b) Find all the x -coordinates where the tangent of the function $f(x) = x^3 - x^2 - 5x + 2$ is parallel to the line $y = 3x + 2$. [4 marks]

(c) Given that $f(x) = \begin{cases} x - 2, & x < 3, \\ \sqrt{x - 2}, & x \geq 3. \end{cases}$

By using the definition $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$, show that f is differentiable at $x = 3$.

[4 marks]

7. A circle and a square are to be formed by using a wire with total length of 8 cm. The radius of the circle is x cm while one of the sides of the square is y cm.

(a) By considering the perimeter of both shapes, find an expression for y in terms of x .
[2 marks]

(b) Show that the total area of two shapes is given by

$$A = \left(\frac{4\pi + \pi^2}{4} \right) x^2 - 2\pi x + 4$$

[3 marks]

(c) If the total area of the two shapes is the minimum possible, what percentage of the wire is used for the circle?
[5 marks]

END OF QUESTION PAPER